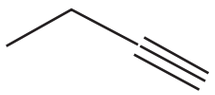
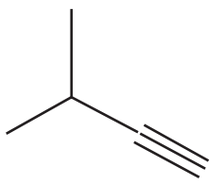
13.4. ALKYNES

Alkynes are unsaturated hydrocarbons that contain carbon-carbon triple bonds in their molecules. Their general formula is C_nH_{2n-2}. The first member of alkyne series is Ethyne popularly known as acetylene. Oxyacetylene torch is used in welding.

Nomencluture of alkynes:

Let us write the IUPAC name for the below mentioned alkynes by applying the general rules of nomenclature that we already discussed in unit No.11

Structure	IUPAC name	Carbon Skeleton Formula
$CH_3—C\equiv CH$	Propyne	
$CH_3\text{-}CH_2—C\equiv CH$	but-1-yne	
$CH_3-C\equiv C-CH_3$	but-2-yne	
$CH_3-CH(CH_3)-C\equiv CH$	3-methyl but-1-yne	

13.4.1. General Methods Of Preparation Of Alkynes

1. Preparation of alkynes from alkenes:

This process involves two steps:

(i) Halogenation of alkenes to form vicinal dihalides

(ii) Dehalogenation of vicinal dihalides to form alkynes.

$$CH_2{=}CH_2 + Br_2 \longrightarrow \underset{\text{1,2-dibromoethane}}{CH_2(Br)\text{-}CH_2(Br)} \xrightarrow{\text{Alcoholic KOH}} \underset{\text{bromoethene}}{CH(Br){=}CH_2} \xrightarrow[-NaBr\ -NH_3]{NaNH_2} \underset{\text{acetylene}}{CH{\equiv}CH}$$

2. Preparation of alkene from gem dihalides:

A compound containing two halogen atoms on the same carbon atom is called gem dihalide (Latin word 'Gemini' means twins). On heating with alcoholic KOH, gem dihalides give alkynes.

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